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noname ~/bit-labs/5th Semester/Information Technology % main ! v15.2.0 14:15
●) g++ lab-1.cpp -o ./build/lab-1 && ./build/lab-1
Enter 1 to encrypt or 2 to decrypt: 1
Enter text (lowercase): top secret message
Enter Key value: 5
Encrypted: ytu xjhwjy rjxxflj

Programmed by Milan Bairagi...
noname ~/bit-labs/5th Semester/Information Technology % main ! v15.2.0 14:16
●) ./build/lab-1
Enter 1 to encrypt or 2 to decrypt: 2
Enter text (lowercase): ytu xjhwjy rjxxflj
Enter Key value: 5
Decrypted: top secret message

Programmed by Milan Bairagi...

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noname ~/bit-labs/5th Semester/Information Technology % main ! v15.2.0 14:20
●) g++ lab-2.cpp -o ./build/lab-2 && ./build/lab-2
Enter 1 to encrypt or 2 to decrypt: 1
Enter text (lowercase): top secret message
Enter substitution key (26 chars, a-z): qwertyuiopasdfghjklzxcvbnm
Encrypted: zgh ltektz dtllqut

Programmed by Milan Bairagi...
noname ~/bit-labs/5th Semester/Information Technology % main ! v15.2.0 14:21
●) ./build/lab-2
Enter 1 to encrypt or 2 to decrypt: 2
Enter text (lowercase): zgh ltektz dtllqut
Enter substitution key (26 chars, a-z): qwertyuiopasdfghjklzxcvbnm
Decrypted: top secret message

Programmed by Milan Bairagi...

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noname ~/bit-labs/5th Semester/Information Technology % main ! v15.2.0 14:31
●) g++ lab-3.cpp -o ./build/lab-3 && ./build/lab-3
Enter 1 to encrypt or 2 to decrypt: 1
Enter text: manoftheword
Enter key: monarchy
Encrypted: orankpcfvdnk

Programmed by Milan Bairagi...
noname ~/bit-labs/5th Semester/Information Technology % main ! v15.2.0 14:31
●) ./build/lab-3
Enter 1 to encrypt or 2 to decrypt: 2
Enter text: orankpcfvdnk
Enter key: monarchy
Decrypted: manoftheword

Programmed by Milan Bairagi...

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noname ~/bit-labs/5th Semester/Information Technology % main ! v15.2.0 14:48
●) g++ lab-4.cpp -o ./build/lab-4 && ./build/lab-4
Enter 1 for encryption and 2 for decryption: 1
===== HILL CIPHER ALGORITHM =====

Key Matrix:
3 3
2 5

Enter plaintext: IAMIRONMAN

Encrypted text: YQIMPAXINN

Programmed by Milan Bairagi...
noname ~/bit-labs/5th Semester/Information Technology % main ! v15.2.0 14:49
●) g++ lab-4.cpp -o ./build/lab-4 && ./build/lab-4
Enter 1 for encryption and 2 for decryption: 2
===== HILL CIPHER ALGORITHM =====

Key Matrix:
3 3
2 5

Enter ciphertext: YQIMPAXINN

Decrypted text: IAMIRONMAN

Programmed by Milan Bairagi...

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```

noname ~/bit-labs % main ! 15:20
●) cd "/home/noname/bit-labs/5th Semester/Information Technology/" && mkdir -p build && g++ lab-5.cpp -o ./build/lab-5 && ./build/lab-5
===== VIGENERE CIPHER ALGORITHM =====

Enter the plaintext: NONAME
Enter the key: WHO

Plaintext: NONAME
Encrypted text: JVBWTS

Ciphertext: JVBWTS
Decrypted text: NONAME

Programmed by Milan Bairagi...

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```

noname ~/bit-labs/5th Semester/Information Security % main ! v15.2.0 16:05
$ g++ lab-6.cpp -o ./build/lab-6 && ./build/lab-6
===== VERNAM CIPHER ALGORITHM =====

Enter the plaintext: SECRETMESAGE

Generated key: UPIQMTQAMJTDD

Plaintext: SECRETMESAGE
Encrypted text: MTKHQMCEEETJH

Ciphertext: MTKHQMCEEETJH
Decrypted text: SECRETMESAGE

Programmed by Milan Bairagi...

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noname ~/bit-labs/5th Semester/Information Security % main ! v15.2.0 13:51
$ g++ lab-7.cpp -o ./build/lab-7 && ./build/lab-7

===== Rail Fence Cipher Algorithm =====
Enter plaintext: HELLO WORLD
Enter number of rails: 3

===== ENCRYPTION =====
Plaintext: HELLO WORLD
Number of rails: 3
Encrypted text: HOREL OLLWD

===== DECRYPTION =====
Ciphertext: HOREL OLLWD
Number of rails: 3
Decrypted text: HELLO WORLD

Programmed by Milan Bairagi...

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```

noname ~/bit-labs % main ! 14:17
$ cd "/home/noname/bit-labs/5th Semester/Information Security/" && mkdir -p build && g++ lab-8.cpp -o ./build/lab-8 && ./build/lab-8

===== Euclidean Algorithm =====
Enter first number: 48
Enter second number: 18

Euclidean Algorithm Steps:
=====
Step | Dividend | Divisor | Remainder | Quotient
-----|-----|-----|-----|-----
1 | 48 | 18 | 12 | 2
2 | 18 | 12 | 6 | 1
3 | 12 | 6 | 0 | 2
=====

GCD(48, 18) = 6

Programmed by Milan Bairagi...

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noname ~/bit-labs/5th Semester/Information Security % main ! v15.2.0 14:43
$ g++ lab-9.cpp -o ./build/lab-9 && ./build/lab-9

===== First 20 Prime Numbers =====
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71

Programmed by Milan Bairagi...

```

```

noname ~/bit-labs/5th Semester/Information Security % main ! v15.2.0 15:05
$ g++ lab-10.cpp -o ./build/lab-10 && ./build/lab-10
Enter a number: 19
19 is prime

Programmed by Milan Bairagi...

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```

noname ~/bit-labs/5th Semester/Information Security % main ! v15.2.0 16:03
$ g++ lab-11.cpp -o ./build/lab-11 && ./build/lab-11
=== Euler's Theorem: Checking Relatively Prime Numbers ===

Enter first number: 9
Enter second number: 11

=== Results ===
9 and 11 are relatively prime.
GCD(9, 11) = 1

=== Verifying Euler's Theorem ===
Euler's Totient  $\phi(11) = 10$ 
 $9^{10} \bmod 11 = 1$ 
Euler's Theorem verified:  $a^{\phi(n)} \equiv 1 \pmod{n}$ 

Programmed by Milan Bairagi...

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```
noname .../bit-labs/5th Semester/Information Security ? main ! v15.2.0 16:25
> g++ lab-12.cpp -o ./build/lab-12 && ./build/lab-12
Enter a number: 11
Numbers coprime to 11: 1 2 3 4 5 6 7 8 9 10
Euler's Totient Function for 11 is 10

Programmed by Milan Bairagi...
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```
noname .../bit-labs/5th Semester/Information Security ? main ! v15.2.0 17:20
> g++ lab-13.cpp -o ./build/lab-13 && ./build/lab-13
=== Diffie-Hellman Key Exchange Algorithm ===
Public Key Generation:
Prime number (P) = 23
Primitive root (G) = 5

Alice's private key (a) = 4
Bob's private key (b) = 3

Alice's public key (x) =  $G^a \text{ mod } P = 4$ 
Bob's public key (y) =  $G^b \text{ mod } P = 10$ 

Alice(x) and Bob(y) exchange their public keys.

Shared secret (Alice):  $y^a \text{ mod } P = 18$ 
Shared secret (Bob):  $x^b \text{ mod } P = 18$ 

Programmed by Milan Bairagi...
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```
noname .../bit-labs/5th Semester/Information Security ? main ! v15.2.0 17:25
> g++ lab-14.cpp -o ./build/lab-14 && ./build/lab-14
=== RSA Algorithm ===

=== Key Generation ===
Chosen prime numbers:
p = 7919
q = 1009
n = p * q = 7919 * 1009 = 7990271
 $\phi(n) = (p-1)(q-1) = 7918 * 1008 = 7981344$ 

Choose public exponent (e) where  $1 < e < \phi(n)$ :
Public exponent (e) = 5
Private exponent (d) = 1596269 (where  $e * d \equiv 1 \pmod{\phi(n)}$ )

=== Public and Private Keys ===
Public Key (e, n) = (5, 7990271)
Private Key (d, n) = (1596269, 7990271)

=== Encryption/Decryption ===
Plaintext message (M) = 123

Encryption:
Ciphertext (C) =  $M^e \text{ mod } n = 123^5 \text{ mod } 7990271 = 3332110$ 

Decryption:
Decrypted Message (M) =  $C^d \text{ mod } n = 3332110^{1596269} \text{ mod } 7990271 = 123$ 

Programmed by Milan Bairagi...
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